

Bruno Avritzer

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U.S Citizen

EDUCATION

University of Southern California, Los Angeles

2024

Ph.D., Physics

Thesis: Quantum and Classical Steganography in Optical Systems

University of Southern California, Los Angeles

2023

M.A., Physics

Relevant course work: Introduction to Quantum Information Processing, Theory of Open Quantum Systems, Quantum Error Correction, Applications of Quantum Computing, Quantum Mechanics 1/2

Cornell University

2019

B.S., Engineering Physics

Relevant course work: Quantum Information Processing, Machine Learning, Data Science Numerics

PROFESSIONAL EXPERIENCE

Leidos, Arlington VA

2024 - present

Quantum Scientist

- Oversaw \$0.5m quantum communications area IRAD portfolio, shaping technical direction in alignment with broader company goals, managing research team of 5, guiding and performing technical work, and preparing and presenting documentation for a nontechnical audience
- Developed, simulated, and presented to DARPA on novel Quantum Secure Direct Communication (QSDC), error correction, channel watermarking, and encoding/decoding protocols for Leidos's optical qudit-based DARPA QuANET offering, revitalizing several lagging project areas while providing an interface between hardware, software, and theory areas for collaborative development
- Shaped engagements and led/PI'd capture efforts from National Security and Defense Sectors
- Fostered cross-sector collaboration in areas as diverse as formal methods and behavioral psychology

University of Southern California, Los Angeles CA

2020 - 2024

Graduate Researcher, Brun Group

- Developed and simulated protocols for covert quantum/classical communication using optical states
- Collaborated with Habif group to demonstrate steganography protocols for the USC DARPA QuANET offering. Developed error correction, encoding, and decoding protocols for weak coherent states

International Business Machines, Littleton MA

2018

Software Development Intern

- Worked with APIC Datapower Support team. Developed/implemented tools to facilitate deployment of API Connect onto servers in the test environment using k8s, HTML, and node.js

SKILLS

Programming languages: Python, Matlab, Java, Javascript, C#, C++, Bash, OCaml

Computer software/ frameworks: Scikit-learn, Pytorch, Numpy, StrawberryFields, Qiskit, Mathematica, Fusion360, AutoCAD, Unity, Arduino, QuTiP, Slurm (HPC)

Languages: English (native), Portuguese (fluent/native), French (basic), Mandarin (basic)

Publications

Clayton, C. & **Avritzer, B.** (2025). Distributed Quantum Error Correction with Permutation-Invariant Approximate Codes. *arXiv 2509.25093*.

Avritzer, B., & Brun, T. (2025). Steganographic Entanglement Sharing. *Phys. Rev. A*, *112*, 022604

Weinstein, H., **Avritzer, B.**, Kinzfohl, C., Brun, T., and Habif, J. (2025). High Fidelity Artificial Quantum Thermal State Generation using Encoded Coherent States. *In IEEE Transactions on Quantum Engineering*, vol. 6, pp. 1-7, 2025, Art no. 5100307.

Avritzer, B., & Brun, T. (2024). Quantum steganography via coherent- and Fock-state encoding in an optical medium. *Phys. Rev. A*, *109*, 032401.

Presentations

Avritzer, B., Wei, A., Clayton, C., and Cochran, R. Connecting and Correcting Multilevel Quantum Systems. *7th International Conference on Quantum Error Correction (QEC25)*.

Avritzer, B., & Brun, T. Steganographic Continuous Variable Entanglement Sharing and Quantum Communication. *APS March Meeting 2024*.

Avritzer, B., & Brun, T. Quantum Steganography in an Optical Channel. *APS March Meeting 2023*.

Avritzer, B., & Brun, T. Quantum Steganography using Coherent states in an Optical Channel. *Southwest Quantum Information and Technology (SQuInT) Workshop 2022*.

Teaching and Mentoring

- USC Jumpstart Program 2020 – provided research mentoring and career development guidance for two junior undergraduates enrolled in USC summer research programs
- Polygence Mentor – provided research mentoring and guidance along with technical help for exceptional high school students. Project title: **Neural Network Decoding of Quantum CSS Codes**
- Leidos Purple Buddy – provided guidance and new hire acclimation for junior employee and connected him to a research group aligned with his career goals
- Teaching Assistant: USC Physics 135b (Physics of Life Sciences), Cornell CS 1132 (Intro to Programming with Matlab), Cornell AEP 3200 (Introductory Mathematical Physics)